

Surya Partap Singh

Senior Data Engineer · Azure · Databricks · Spark · Lakehouse Platforms

psurya.924@gmail.com | linkedin.com/in/surya6032 | github.com/Surya6032

SUMMARY

- Senior Data Engineer with 5+ years building and scaling enterprise lakehouse platforms on **Databricks, Delta Lake, Apache Spark, and Azure** — unifying clinical, financial, and pharmacy data for analytics, reporting, and ML. Proven track record of optimizing distributed workloads, automating CI/CD pipelines, and mentoring engineering teams in high-stakes production environments.

TECHNICAL SKILLS

- **Programming:** Python, PySpark, Spark SQL, SQL, C++
- **Data Platforms:** Databricks, Delta Lake, Apache Spark, Hadoop, Hive, HDFS
- **Cloud & Storage:** Microsoft Azure (Data Lake, Blob Storage, Azure SQL, ADF), Amazon S3
- **Streaming:** Spark Structured Streaming, Delta Change Data Feed
- **Orchestration:** Databricks Workflows, Azure Data Factory, Airflow
- **Data Quality & Governance:** Great Expectations, schema evolution, anomaly detection, data lineage, observability
- **DevOps & IaC:** GitHub Actions, Azure DevOps, ARM Templates, Git, Docker
- **Ways of Working:** Agile, code reviews, mentoring, production support

EXPERIENCE

- **Cencora** Dallas, TX
Senior Data Engineer, Data Platform *Feb 2023 – Present*
 - **Lakehouse Platform Architecture:** Designed and scaled a **Databricks** and **Delta Lake**-based medallion lakehouse that unified clinical, financial, and pharmacy data into reusable, governed datasets — reducing time-to-insight for analytics, reporting, and data science teams by **50%**.
 - **Spark Performance Optimization:** Improved end-to-end pipeline throughput by **30%** by tuning **Apache Spark** workloads across partitioning, shuffle behavior, join strategies, and cluster resource utilization.
 - **Near-Real-Time Processing:** Reduced financial reporting latency from **24 hours to 15 minutes** by redesigning batch pipelines into incremental and near-real-time processing using **Delta CDF** and **Spark Structured Streaming**.
 - **Unstructured Data Pipelines:** Built OCR-driven ingestion pipelines to extract and standardize data from unstructured patient documents, enabling curated datasets for clinical analytics and downstream ML workflows.
 - **Data Quality & Observability:** Implemented automated data quality controls using **Great Expectations** and anomaly detection frameworks — catching schema drift, freshness failures, and integrity issues before downstream impact, reducing data incidents by **35%**.
 - **Platform Automation:** Established CI/CD and infrastructure automation for data pipelines and cloud resources using **GitHub Actions**, **Azure DevOps**, and **ARM Templates**, improving release consistency and cutting deployment time by **25%**.
 - **Reusable Frameworks:** Developed reusable ingestion and transformation frameworks along with canonical data models, reducing onboarding time for new data sources by **40%**.
 - **Technical Leadership:** Mentored a team of **3–5 engineers** on ETL design, Spark optimization, code quality, and platform best practices while influencing standards for scalable data engineering.
- **Cencora (Contract)** Dallas, TX
Data Engineer *May 2021 – Feb 2023*
 - **Production Data Pipelines:** Built and supported production ETL pipelines on **Databricks**, **Azure Data Factory**, and **PySpark** for large-scale clinical, pharmacy, and financial datasets.
 - **Canonical Data Models:** Designed canonical, single-source-of-truth tables that standardized disparate source systems — improving cross-team reporting consistency and eliminating **3+** redundant dataset copies maintained across business units.

- **Distributed Data Processing:** Optimized large-scale **Spark** transformations using broadcast joins, predicate pushdown, and partition-aware reads to improve runtime efficiency and reduce compute costs.
- **Workflow Reliability:** Automated and monitored ETL workflows through **Databricks Jobs**, implementing retries, alerting, and operational checks — improving pipeline success rates to **99%+** and ensuring consistent SLA adherence.

- **Fuge Technologies Inc**

Dallas-Fort Worth, TX

Data Engineer

Feb 2021 – May 2021

- **Batch Processing:** Built and optimized Hadoop and Hive-based batch pipelines for large structured datasets, applying partition pruning, bucketing strategies, and execution plan analysis to improve query performance in distributed environments.
- **Azure Data Ingestion:** Developed ingestion pipelines into Azure Data Lake using **Azure Data Factory**, supporting incremental loads, schema evolution handling, and downstream validation checks for analytics use cases.
- **Data Standardization:** Collaborated with cross-functional teams to normalize and standardize raw source data into consistent, analytics-ready formats, establishing naming conventions and structural patterns adopted across the pipeline.

EDUCATION

- **University of North Alabama**

Florence, AL

Bachelor of Science in Computer Science

Aug 2016 – Dec 2020

- **Relevant Coursework:** Distributed Systems, Database Management Systems, Data Structures & Algorithms, Operating Systems, Cloud Computing, Software Engineering, Machine Learning